

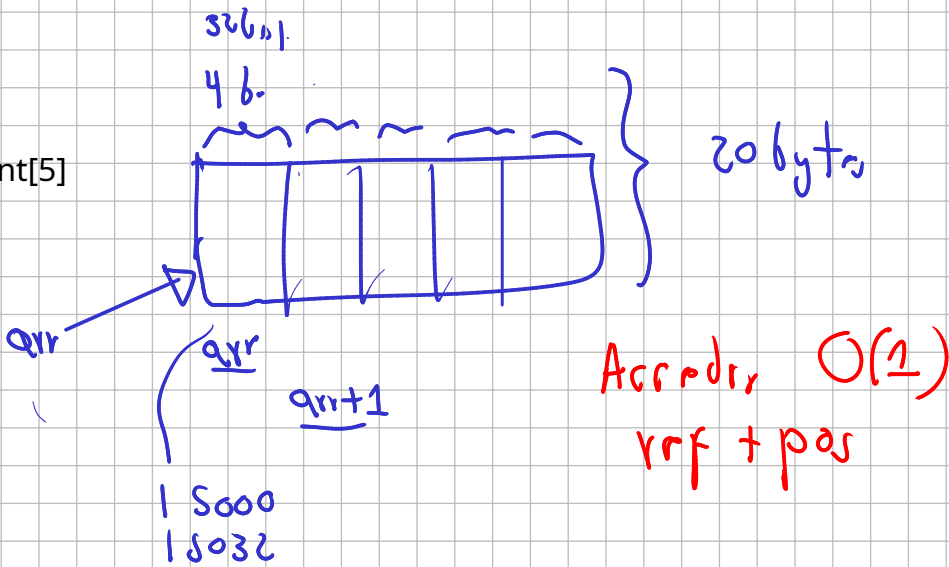
Estructuras de datos son mecanismos que nos permiten almacenar información, las estructuras nos proveen operaciones

- Lineales: Arreglos, Listas enlazadas
- I/O: Cola, Pila
- Map: Hashmap (Tablas hash)
- Arboles

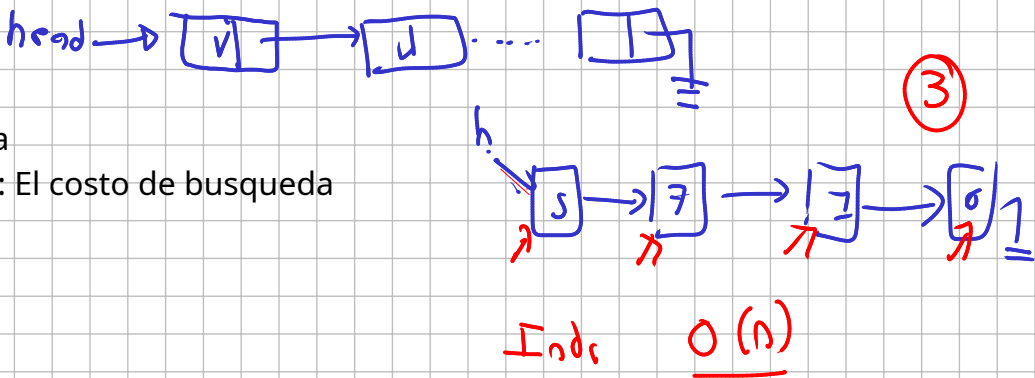
Arreglos: Son una reserva de espacio de memoria, usualmente son tipados (espacio ocupar)

- integer: 32 bits
- long: 64 bits
- double: 64 bits
- float: 32 bits
- short: 8 bits
- char: 8 bits
- byte: 8 bits

```
int* arr = new int[5]
```



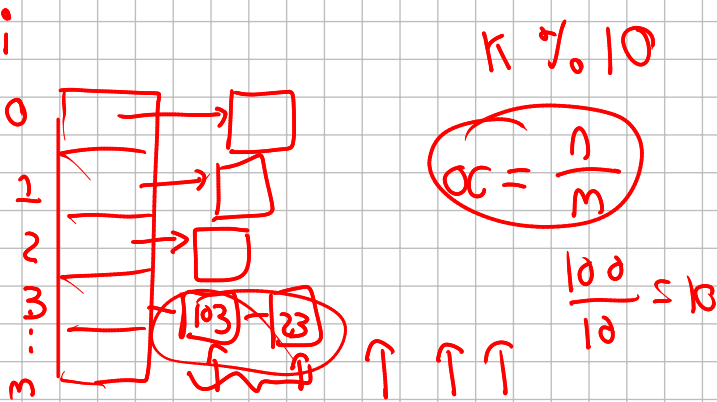
Lista enlazada



Es dinamica
Desventaja: El costo de busqueda

Tabla hash

(k, v)
 $103 \% 10 = 3$
 $23 \% 10 = 3$
 $h(k) = i$



Costo de busqueda

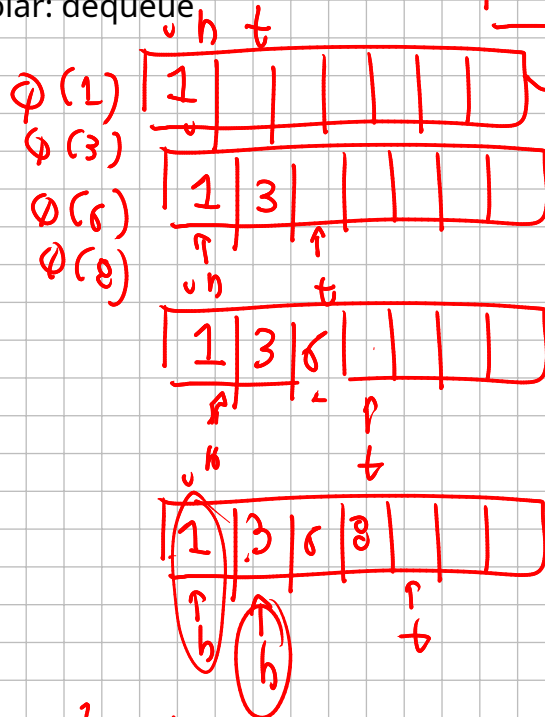
k_1
1) $k_1 \% m \rightarrow i$ $O(1)$
2) $O(\alpha)$
} $O(1 + \alpha)$

Queue / Cola

FIFO

Esta estructura tiene las operaciones

- Encolar: enqueue
- Desencolar: dequeue

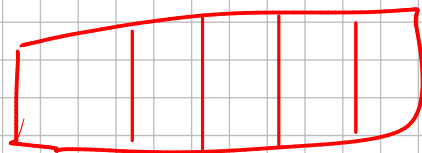


$$arr[t] = v \quad O(1)$$

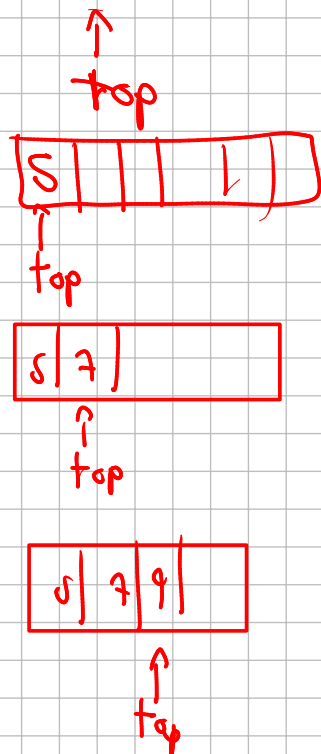
$$O(1) \rightarrow \begin{matrix} h+1 & O(1) \\ arr[h] & O(1) \\ \hline & O(1) \end{matrix}$$

Stack / Pila

LIFO



push
pop



push(s)

push(7)

push(9)

$$\left. \begin{matrix} top++ \\ arr[top] = v \end{matrix} \right\} O(1)$$

pop()

arr[top]

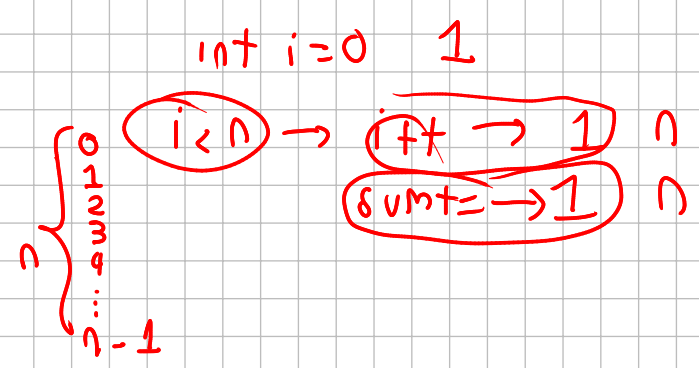
top--

$$\left. \begin{matrix} arr[top] \\ top-- \end{matrix} \right\} O(1)$$

```

int suma = 0;
for(int i = 0; i < n; i++){
    suma += est[i]
}

```



$1 + 1 + (n+1) + n + n \rightarrow 3n + 3 \rightarrow O(n)$
int i = 0 i < n i++ i++

lista
 pot → enb zqda

arr[0] = 1
 arr[1] = 2
 2 → 3
 :
 n-1 → n

$\sum_{i=1}^n i = \frac{n(n+1)}{2}$

$1 + 1 + (n+1) + n + \frac{n(n+1)}{2}$

$\frac{4n+6}{2} + \frac{n(n+1)}{2} = \frac{4n+6+n^2+n}{2} \rightarrow O(n^2)$

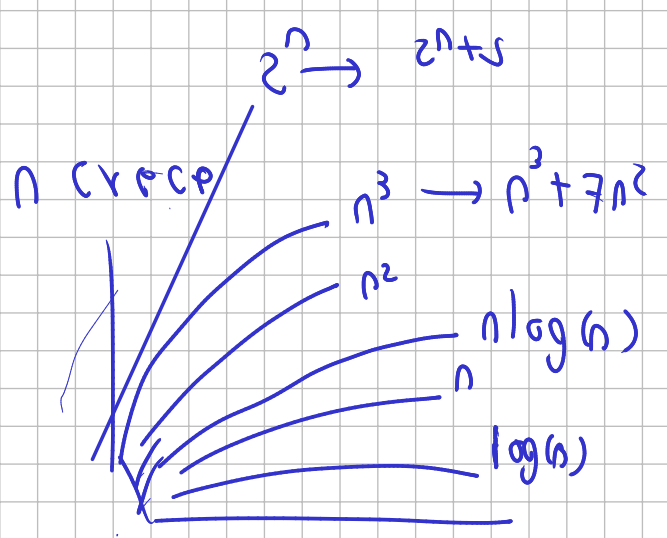
1 ————— 100
 1 + 100 = 101
 2 + 99 = 101
 3 + 98 = 101
 } so vres $\left(\frac{n}{2}\right)(n+1)$

for i in lista

for i in range(0,n)
 lista[i]

Notacion asintotica

$2n+8$
 n^2+7n+5

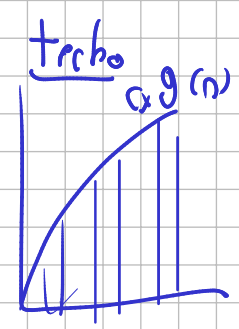


$O(n)$

$f(n) \in O(g(n))$ si

$f(n) \leq c \times g(n) \quad \forall n \geq k$

$n^2 \in O(n^3)$
 $7n^3+8n \in O(n^3)$



$n^3 \in O(n^2)$
NO

2

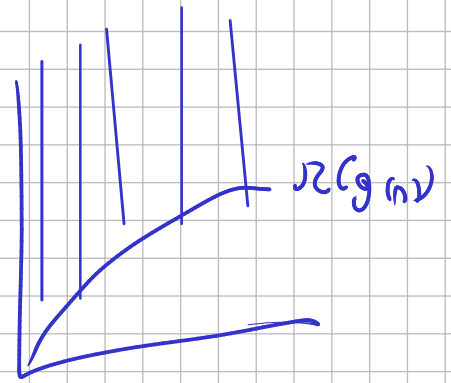
$$f(n) \approx \Omega(g(n))$$

$$f(n) \geq c \cdot g(n) \quad n \geq k$$

$$n^2 \approx \Omega(n^2)$$

$$n^3 \approx \Omega(n^2) \checkmark$$

Major
Caso



(+) Coste ajustado

$$f(n) \approx \Theta(g(n))$$

$$f(n) \approx O(g(n)) \text{ y } \Omega(g(n))$$

$$n^2 + n \approx \underline{\Theta(n^2)}$$

$$3n^2 - n \approx \underline{\Theta(n^2)}$$

Establecido

$$n + 1 \approx \underline{\Theta(n)}$$

$$1 + n^2 \approx \underline{\Theta(1)}$$

$O(1)$ X

$\Omega(1)$ \checkmark